

# Waterloo Quant Club

## Spring 2026 Trading Competition Packet

### Contents

|                                     |           |
|-------------------------------------|-----------|
| <b>Introduction</b>                 | <b>3</b>  |
| <b>Competition Format</b>           | <b>3</b>  |
| <b>Overall Scoring Format</b>       | <b>3</b>  |
| Overview . . . . .                  | 3         |
| Tiebreakers . . . . .               | 4         |
| Motivation . . . . .                | 4         |
| <b>ETF Trading Frenzy</b>           | <b>5</b>  |
| Overview . . . . .                  | 5         |
| Game Structure . . . . .            | 5         |
| Symbols and Settlement . . . . .    | 5         |
| Customer and Bot Behavior . . . . . | 6         |
| Scoring . . . . .                   | 6         |
| <b>Octomarket</b>                   | <b>7</b>  |
| Overview . . . . .                  | 7         |
| Game Structure . . . . .            | 7         |
| Symbols and Settlement . . . . .    | 7         |
| Information Release . . . . .       | 8         |
| Customer and Bot Behavior . . . . . | 8         |
| Scoring . . . . .                   | 8         |
| <b>Triplet of Pairs</b>             | <b>9</b>  |
| Overview . . . . .                  | 9         |
| Game Structure . . . . .            | 9         |
| Symbols and Settlement . . . . .    | 9         |
| Event Mechanism . . . . .           | 9         |
| Information Release . . . . .       | 9         |
| Customer and Bot Behavior . . . . . | 10        |
| Scoring . . . . .                   | 10        |
| <b>Suit Factions</b>                | <b>11</b> |
| Overview . . . . .                  | 11        |

|                                   |           |
|-----------------------------------|-----------|
| Game Structure . . . . .          | 11        |
| Trading Rounds . . . . .          | 11        |
| Government Shop . . . . .         | 12        |
| Faction Abilities . . . . .       | 12        |
| Bluffing and Challenges . . . . . | 13        |
| Scoring . . . . .                 | 13        |
| <b>Infinite Auction</b>           | <b>15</b> |
| Overview . . . . .                | 15        |
| Game Structure . . . . .          | 15        |
| Bidding Rounds . . . . .          | 15        |
| Round Resolution . . . . .        | 15        |
| Information Release . . . . .     | 15        |
| Scoring . . . . .                 | 16        |

## Introduction

Welcome to the Waterloo Quant Club (WQC) Spring 2026 Trading Competition. Over the course of the event, you will play through five games designed to test different aspects of quantitative trading and strategic decision-making, including market making, trading under uncertainty, information aggregation, auctions, and negotiation.

The competition will feature five main games:

- **ETF Trading Frenzy:** live market making and trading in single names and an ETF whose settlement depends on aggregate trading flow.
- **Octomarket:** trading binary-style contracts on statistics of two simulated teams' dice rolls.
- **Triplet of Pairs:** trading correlated contracts using staggered private information about hidden sampling distributions.
- **Suit Factions:** a social trading and negotiation game built around cards, hidden rules, faction abilities, and government interaction.
- **Infinite Auction:** repeated all-pay style bidding with point rewards, penalties, and bid-based endgame adjustments.

Your final standing in the competition will be determined by how well your teams place across these five games, using a geometric-mean based scoring system described below.

Beyond the scoreboard, the goal of the day is to give participants a hands-on feel for the kinds of thinking that matter in quantitative trading: pricing under uncertainty, reacting to order flow, reasoning from partial information, allocating resources dynamically, and coordinating effectively as a team.

## Competition Format

The competition is scored at the **individual** level. Each game is scored at the **team** level.

- Participants are placed in different teams across games.
- If a game is played in multiple rounds, your team accumulates performance across those rounds before being ranked in that game.
- The game ranking received by a team is shared by all members in that team, which is used to calculate their individual scores for the competition.
- Failure to submit, connect, or participate in a required action in a game may be treated as a last-place finish for that component of the game.

## Overall Scoring Format

### Overview

Each game produces a placement for every team: first place, second place, third place, and so on. Let the placements of your teams across the five scored games be

$$p_1, p_2, p_3, p_4, p_5,$$

where 1 = first place and larger numbers are worse.

Your overall competition score is defined as the geometric mean of your placements:

$$S = \left( \prod_{r=1}^5 p_r \right)^{1/5}.$$

Individuals are ranked overall by *minimizing*  $S$ . Smaller values correspond to stronger and more consistent performance across the competition.

## Tiebreakers

Ties on the overall leaderboard will be broken by, in order:

1. Lower arithmetic mean of placements,

$$\frac{1}{5} \sum_{r=1}^5 p_r.$$

2. Best single-game placement.
3. Random tiebreaker if still tied.

## Motivation

This scoring system rewards individuals that play to win while still remaining robust to one poor result:

- A very strong finish in a game meaningfully improves your overall standing.
- One bad game hurts, but is not fatal.
- Every game matters, since each enters multiplicatively into the final score.

# ETF Trading Frenzy (E.T.F.)

## Overview

*ETF Trading Frenzy* is a live electronic trading game featuring the three underlying symbols  $\mathbb{E}$ ,  $\mathbb{T}$ , and  $\mathbb{F}$ , and one special symbol,  $\mathbb{ETF}$ . The settlement of  $\mathbb{ETF}$  depends on the eventual settlements of the underlyings. Teams will trade these assets in the presence of noise flow, event-driven flow, and passive making bots.

## Game Structure

- The game is played in teams of **3 to 4 people**. Teams are formed randomly and remain fixed across rounds.
- The game consists of **3 rounds** in total, each of which lasts **10 minutes**.
- In each round, teams are assigned to exchanges, with **5 to 6 teams** per exchange. Exchanges are shuffled between rounds, so teams face different opponents each round.
- Orders in the exchange follow the below specifications:
  1. The minimum price variation is \$1.
  2. Prices will remain within a band from \$300 to \$700. This means teams will not be able to submit orders with prices outside of this band.
  3. Teams may only have 10 active orders in each symbol at any time.

## Symbols and Settlement

The traded symbols are

$$\mathbb{E}, \quad \mathbb{T}, \quad \mathbb{F}, \quad \mathbb{ETF}.$$

Each of the three single-name assets,  $\mathbb{E}$ ,  $\mathbb{T}$ , and  $\mathbb{F}$ , settles based on the *net trade volume* generated by (and thus, the finishing position of) the *noise bots* and *event bots* (covered later). Specifically, for any symbol  $X \in \{\mathbb{E}, \mathbb{T}, \mathbb{F}\}$ , let  $P_{\text{NB}}(X)$  and  $P_{\text{EB}}(X)$  be the finishing position in shares of the noise bot and event bot respectively in  $X$ . Then the settlement of  $X$  is given by

$$\text{Settlement}(X) = 500 + \frac{P_{\text{NB}}(X) + P_{\text{EB}}(X)}{100}, \quad X \in \{\mathbb{E}, \mathbb{T}, \mathbb{F}\}.$$

The symbol  $\mathbb{ETF}$  settles as

$$\text{Settlement}(\mathbb{ETF}) = \frac{\mathbb{E} + 2\mathbb{T} + 3\mathbb{F}}{6}.$$

## Customer and Bot Behavior

During trading, there will be bots present in the market that generate various types of flow. The different types of bots are described below.

- **Noise Bots** are present in each of  $\mathbb{E}$ ,  $\mathbb{T}$ ,  $\mathbb{F}$ , and  $\mathbb{ETF}$ . They continuously fill bids and offers throughout the round. The fill size is relative to the market width; tighter markets induce larger noise bot sizes. Fill sizes are also capped at the best bid/offer sizes for each direction.
- **Event Bots** create events signified by bursts of directional trading in one symbol.
  1. For each event, the target symbol (one of  $\{\mathbb{E}, \mathbb{T}, \mathbb{F}\}$ ) and trade direction (buy or sell) are both selected randomly with uniform probability.
  2. Once selected, the bots send a large sequence of fills in quick succession to the symbol in the chosen direction.
  3. The fill sizes are relative to the market width; tighter markets induce larger event bot sizes.
- **Making Bots** are present in each of  $\mathbb{E}$ ,  $\mathbb{T}$ ,  $\mathbb{F}$ , and  $\mathbb{ETF}$ . They provide passive liquidity by continuously posting resting limit orders on both sides of the market.

## Scoring

- In each round, teams are ranked based on their marked-to-settlement PnL for that round.
- At the end of all rounds, teams are ranked within the game based on the **arithmetic mean of all round placements**. In the case of ties, it is broken using the teams' **cumulative PnL** across all rounds.
- The game ranking determines your placement  $p_1$  for the overall competition score.

# Octomarket

## Overview

*Octomarket* is an event-driven prediction market game. Two fictional teams,  $A$  and  $B$ , play a simulated game consisting of 10 independent d20 (20-sided die) rolls each. Trading teams observe the realized rolls over time and may trade several binary contracts whose payoffs depend on various attributes of the completed roll sequences.

## Game Structure

- The game is played in teams of **3 to 4 people**. Teams are formed randomly and remain fixed across rounds.
- The game consists of **2 rounds** in total, each of which lasts **11 minutes**.
- In each round, teams are assigned to exchanges, with **5 to 6 teams** per exchange. Exchanges are shuffled between rounds, so teams face different opponents each round.
- Orders in the exchange follow the below specifications:
  1. The minimum price variation is \$1.
  2. Prices will remain within a band from \$0 to \$100. This means teams will not be able to submit orders with prices outside of this band.
  3. Teams may only have 10 active orders in each symbol at any time.

## Symbols and Settlement

Let  $A_1, \dots, A_{10}$  and  $B_1, \dots, B_{10}$  denote the realized die rolls for teams  $A$  and  $B$ , respectively. Define the sums and products  $S_A = \sum_{i=1}^{10} A_i$ ,  $S_B = \sum_{i=1}^{10} B_i$ ,  $P_A = \prod_{i=1}^{10} A_i$ ,  $P_B = \prod_{i=1}^{10} B_i$ . The traded symbols are

WIN\_A, WIN\_B, ODD\_A, ODD\_B, DIV\_A, DIV\_B, SPREAD\_S, SPREAD\_L.

A description of the settlement conditions for each symbol is given below.

- **Winner contracts:** WIN\_A settles to 100 if  $S_A$  is greater than  $S_B$ , 50 if tied, and 0 otherwise. WIN\_B settles to 100 if  $S_B$  is greater than  $S_A$ , 50 if tied, and 0 otherwise.
- **Odd-count contracts:** ODD\_A settles to 100 if the number of odd  $A_i$  is greater than the number of odd  $B_i$ , 50 if tied, and 0 otherwise. ODD\_B settles to 100 if the number of odd  $B_i$  is greater than the number of odd  $A_i$ , 50 if tied, and 0 otherwise.
- **Divisor-count contracts:** DIV\_A settles to 100 if the number of divisors of  $P_A$  is greater than the number of divisors of the product of  $P_B$ , 50 if tied, and 0 otherwise. DIV\_B settles to 100 if the number of divisors of  $P_B$  is greater than the number of divisors of  $P_A$ , 50 if tied, and 0 otherwise.
- **Spread contracts:** SPREAD\_S settles to 100 if  $|S_A - S_B| > 5.5$ , and 0 otherwise. SPREAD\_L settles to 100 if  $|S_A - S_B| > 20.5$ , and 0 otherwise.

## Information Release

All trading teams observe the same public sequence of rolls, which is revealed every minute.

1. At 1 minute, rolls  $A_1$  and  $B_1$  are revealed.
2. At 2 minutes, rolls  $A_2$  and  $B_2$  are revealed.
- ...
10. At 10 minutes, rolls  $A_{10}$  and  $B_{10}$  are revealed.

## Customer and Bot Behavior

During trading, there will be bots present in the market that generate various types of flow. The different types of bots are described below.

- **Noise Bots** are present in each of the symbols traded. They continuously fill bids and offers throughout the round. The fill size is relative to the market width; tighter markets induce larger noise bot sizes. Fill sizes are also capped at the best bid/offer sizes for each direction.
- **Direction Bots** are present in each of the symbols traded. They are similar to noise bots, but will only buy (i.e. lift offers). The fill size is relative to the market width; tighter markets induce larger direction bot sizes. Fill sizes are also capped at the best offer sizes.
- **Making Bots** are present in each of the symbols traded. They provide passive liquidity by continuously posting resting limit orders on both sides of the market.

## Scoring

- In each round, teams are ranked based on their marked-to-settlement PnL for that round.
- At the end of all rounds, teams are ranked within the game based on the **arithmetic mean of all round placements**. In the case of ties, it is broken using the teams' **cumulative PnL** across all rounds.
- The game ranking determines your placement  $p_3$  for the overall competition score.



# Triplet of Pairs

## Overview

*Triplet of Pairs* is a correlated-value trading game involving three hidden events:  $\mathbb{A}$ ,  $\mathbb{B}$ , and  $\mathbb{C}$ .

Each event is generated using repeated draws from a d5 (a 5-sided die) with initially unknown sides. Teams trade contracts on the pairwise sums of the three events, while receiving staggered information about exactly one of the events. Teams will also trade against various bots which provide extra liquidity.

## Game Structure

- The game is played in teams of **2 to 3 people**. Teams are formed randomly and remain fixed across rounds.
- The game consists of **3 rounds** in total, each of which lasts **10 minutes**.
- In each round, teams are assigned to exchanges, with **6 teams** per exchange. Exchanges are shuffled between rounds, so teams face different opponents each round.
- Orders in the exchange follow the below specifications:
  1. The minimum price variation is \$1.
  2. Prices will remain within a band from \$18 to \$180. This means teams will not be able to submit orders with prices outside of this band.
  3. Teams may only have 10 active orders in each symbol at any time.

## Symbols and Settlement

The traded symbols are

$$\mathbb{AB}, \quad \mathbb{BC}, \quad \mathbb{AC}.$$

These contracts settle based on the realized values of hidden events  $\mathbb{A}$ ,  $\mathbb{B}$ ,  $\mathbb{C}$ :

$$\mathbb{AB} = \mathbb{A} + \mathbb{B}, \quad \mathbb{BC} = \mathbb{B} + \mathbb{C}, \quad \mathbb{AC} = \mathbb{A} + \mathbb{C}.$$

## Event Mechanism

Each of the events  $\mathbb{A}$ ,  $\mathbb{B}$ ,  $\mathbb{C}$  is resolved independently as follows:

1. Form the faces of a d5 by drawing independently and **with replacement** from  $\{1, 2, \dots, 10\}$ .
2. A total of 9 integers are sequentially rolled **from the same d5**.
3. The realized event value is the sum of the 9 integers rolled.

## Information Release

Each team is assigned to one of  $\mathbb{A}$ ,  $\mathbb{B}$ ,  $\mathbb{C}$ , and receives information only for that event. Information is released every 20 seconds in *cyclic order*:

1. At 20 seconds, teams assigned to  $\mathbb{A}$  receive the result of the first roll for event  $\mathbb{A}$ .

2. At 40 seconds, teams assigned to  $\mathbb{B}$  receive the result of the first roll for event  $\mathbb{B}$ .
3. At 1 minute, teams assigned to  $\mathbb{C}$  receive the result of the first roll for event  $\mathbb{C}$ .
- ...
26. At 8 minutes and 40 seconds, teams assigned to  $\mathbb{B}$  receive the result of the ninth and final roll for event  $\mathbb{B}$ .
27. At 9 minutes, teams assigned to  $\mathbb{C}$  receive the result of the ninth and final roll for event  $\mathbb{C}$ .

## Customer and Bot Behavior

During trading, there will be bots present in the market that generate various types of flow. The different types of bots are described below.

- **Noise Bots** are present in each of  $\mathbb{AB}$ ,  $\mathbb{BC}$ , and  $\mathbb{AC}$ . They continuously fill bids and offers throughout the round. The fill size is relative to the market width; tighter markets induce larger noise bot sizes. Fill sizes are also capped at the best bid/offer sizes for each direction.
- **Making Bots** are present in each of  $\mathbb{AB}$ ,  $\mathbb{BC}$ , and  $\mathbb{AC}$ . They provide passive liquidity by continuously posting resting limit orders on both sides of the market.

## Scoring

- In each round, teams are ranked based on their marked-to-settlement PnL for that round.
- At the end of all rounds, teams are ranked within the game based on the **arithmetic mean of all round placements**. In the case of ties, it is broken using the teams' **cumulative PnL** across all rounds.
- The game ranking determines your placement  $p_2$  for the overall competition score.

# Suit Factions

Preface: This is by far the most complex game rules-wise. We recommend participants to carefully familiarize themselves with the rules in advance, and to direct inquiries through Discord or email.

## Overview

Suit Factions is a live trading, negotiation, and information game played with cards and tokens. Each team begins with a hand of cards and a token endowment. A team's faction is determined by the suit composition of its hand, and each faction has its own rules and abilities. Over five periods, teams trade cards and tokens, purchase information and assets from a government, and attempt to assemble a hand that scores well under gradually revealed faction-dependent rules.

## Game Structure

Each team starts with:

- 10 playing cards,
- 200 tokens,
- and for each  $r \in \{1, 2, 3, 4, 5\}$  and  $s = \{\text{Spade, Heart}\}$ , 2 slips of paper with their team number and the description "Round  $r$ ,  $s$  Faction Ability" (total 20 slips).

This section and the rules to be presented on the competition day will make frequent mentions of a card's *rank*. We define this as the face value of the card, with the following special cases:

$$\text{Ace} = 1 \quad \text{Jack} = 11 \quad \text{Queen} = 12 \quad \text{King} = 13$$

A team's **faction** is determined as follows:

1. The most frequent suit in their hand determines their faction.
2. If tied between multiple suits, the tying most frequent suit with the highest card(s), where ties in the highest  $n$  cards are broken by the  $(n + 1)$ -th card's rank.
3. If still tied, then by the suit order  $\spadesuit > \heartsuit > \clubsuit > \diamondsuit$ .

Each suit faction has:

- 7 faction-specific hand scoring rules,
- a faction **ability**.

## Trading Rounds

There are **5 total trading rounds**, each lasting **8 minutes**. At the start of each round, one rule is publicly revealed for each suit. All faction abilities are known at the start of the game.

During active trading rounds, teams may trade cards, tokens, and information with each other, subject to the following:

- Before a trade, teams may negotiate freely.
- It is the responsibility of both teams to clearly outline the terms of a trade. These terms cannot be broken, except using the Heart Ability, Hidden Swap.

- To prevent excessive and unfair information gain during negotiations, teams are not allowed to show any card in their possession to another team. As a result, trades must be simultaneous. During a trade, the relevant cards should be placed face down simultaneously, and a team is only allowed to view the cards they requested after the trade has completed.
- Teams may pay for information from other teams, though such information need not be truthful.
- During negotiations only, a team must truthfully disclose whether or not they have 30 or more tokens if requested, but they do not need to disclose the exact amount.

## Government Shop

Teams may transact with the government, which offers goods in exchange for tokens:

- **Private Rules:** A team will be shown a private rule of their choice. [Cost: 50 Tokens]
- **Future Public Rules:** A team will be shown an unrevealed public rule of their choice that is planned to reveal in a future round. [Cost:  $10 \times N$  Tokens, where  $N$  is the amount of rounds away from the rule's exposure]
- **Random Card:** A team is given a random playing card. [Cost: 15 Tokens]
- **Suited Card:** A team is given a random playing card of a particular suit they may specify. [Cost: 25 Tokens]
- **Any Card:** A team is given a playing card of their choice. [Cost: 50 Tokens]

During the competition, there will be various government booths stationed around the venue.

## Faction Abilities

Faction abilities alter trading but do not directly affect the score of a hand.

**Spade Ability, Ultimatum:** You may approach a team and demand for a certain card (specify rank and suit) from them. They can choose to give you that card. If they choose not to, then you may **either** steal 30 tokens (or their remaining tokens, if insufficient), or a random card from their hand. A team may use this ability up to twice per round, but only once against a specific opposing team each round.

**Heart Ability, Hidden Swap:** During a negotiated trade, you may replace all cards you are offering up with different cards. **Following the deceptive trade, you must notify the opposing team of the swap.** A team may use this ability up to twice per round, but only once against a specific opposing team each round.

**Club Ability, Insurance:** When Ultimatum or Hidden Swap is used on your team, you may nullify its effect, and then **either** steal 20 tokens (or their remaining tokens, if insufficient), or a random card from their hand. This does not count as an ability usage for the opposing team. There is no limit on the amount of times this ability can be used per round, but it can only be used in reaction to an opposing team using Ultimatum or Hidden Swap.

**Diamond Ability, Well-Connected:** Every round, you may collect 40 tokens from the government, and all rule purchases will cost half price. You also gain the ability to buy tradeable Point Cards from the government for 10 tokens each; these give a team 1 raw score regardless of their faction.

**Ability Usage Logistics:** When your team successfully uses Ultimatum or Hidden Swap, you should present one of your slips indicating your team number, the current round number, and appropriate ability to the other team. This exchange both ensures that the opposing team has a certificate that your team has already used a certain ability (and cannot use the same ability again) on them this round, and that you have 1 fewer slips this round to give out, limiting your ability usages.

## Bluffing and Challenges

Teams with fewer than 30 tokens must prove they are from the appropriate faction when using a faction ability, by showing enough cards such that it is certain they are of that faction.

For example, if I want to use Ultimatum and I have 11 cards, I can show 6 spades and my remaining 5 face-down cards to undeniably prove I am from the spade faction.

Teams with 30 tokens or more may use any faction ability without proving they are from that faction, with the exception of Well-Connected (since this ability only affects interactions with the government). However, the opposing team is given a chance to **challenge** the ability usage:

- If the opposing team does not challenge, the ability proceeds as usual.
- if the opposing team challenges, your team must prove it was entitled to use the ability in a similar manner to teams with fewer than 30 tokens using abilities.
  - If your team is able to prove legitimacy, the opposing team pays you 30 tokens, and the ability proceeds as usual. If they have fewer than 30 tokens, they must give you their remaining tokens.
  - If your team is unable to provide proof, you must pay the opposing team 30 tokens, and the ability does not proceed. This does not count as an ability usage for your team.
- In the special case of the Hidden Swap ability, your team will only become aware of its usage following the trade. If your team wants to challenge, you should temporarily return the traded cards back to each other first before the opposing team proves legitimacy (we recommend keeping the cards involved in the trade separate from each team's hands, such as face-up on a table). This way, the opposing team is allowed to use the swapped cards they traded away as proof, but not the cards you gave them (effectively their pre-trade hand). If the opposing team is able to provide proof, restore the deceptive trade; otherwise do not.
- Against a usage of Ultimatum or Hidden Swap, you must first decide whether you want to challenge before you may use Insurance. If you successfully challenge an Ultimatum or Hidden Swap usage, you cannot then use Insurance.

## Scoring

A team's **final score** is computed as follows:

- The team's faction is determined, and a **raw score** is computed using that faction's rules. It is guaranteed that rules will not interfere with other rules.
- The faction with the highest mean raw score gains 20 extra points for all its members. The faction with the lowest mean raw score loses 20 points for all its members. In the case of a tie, the point bonus or penalty is split between all teams that tied, rounded down.

- The faction with the least members gains 20 extra points for all its members. The faction with the most members loses 20 points for all its members. In the case of a tie, the point bonus or penalty is split between all teams that tied, rounded down.

Teams are ranked within Suit Factions by *higher* final score. That ranking determines your placement  $p_5$ .

# Infinite Auction

## Overview

Infinite Auction is a repeated sealed-bid game involving 24 teams over 20 rounds. In each round, teams bid for points. The highest bidders win points, while the lowest bids are penalized. However, bidding also affects scoring through a bid-based penalty applied at the end of the game.

## Game Structure

- The game is played in teams of **4 to 6 people**. Teams are formed randomly and remain fixed across rounds.
- The game consists of **20 bidding rounds** in total.
- All teams will participate in the same instance of the game.

## Bidding Rounds

In each round, anywhere between 1 to 5 points will be up for auction. The sequence of point values across the 20 rounds will be fixed and announced in advance.

During an auction, every team has 60 seconds to submit exactly one bid, consisting of a nonnegative integer not exceeding 100,000,000. Failure to submit a valid bid will result in a bid of 0 being assigned. Teams have no budget and may bid as much as they want, but must be cautious of the cumulative bid penalties.

**Point Multiplier:** Before submitting a bid, a team may optionally purchase a *Point Multiplier* for this round for a cost of 0.5 points. If purchased, the auction prize for this team only, and for this round only, will be doubled.

## Round Resolution

After all bids have been submitted, winners are determined as follows:

- Let  $B_3$  be the third highest (possibly duplicate) bid if all bids are put in sorted order.
- All teams with bid  $\geq B_3$  will win the points.

There are penalties for bidding too little in a round:

- Let  $B_{-3}$  be the third lowest (possibly duplicate) bid if all bids are put in sorted order.
- All teams with bid  $\leq B_{-3}$  will lose 1 point. Point Multipliers do not affect this penalty.

## Information Release

Various public information will be revealed throughout the game:

- After every round, a full leaderboard of cumulative points is displayed. The end-of-game bid penalty is **not** reflected here.
- After every 4 rounds, the following statistics for the preceding block of 4 rounds are revealed:
  - the highest bid in each of the rounds,

- the lowest winning bid in each of the rounds.

## Scoring

For each team, let:

- $P$  = cumulative point total after all round rewards, penalties, and multiplier effects,
- $B$  = total sum of bids submitted across all 20 rounds. Importantly, note that **even if you do not win a round, your bids still contribute to  $B$ .**

Define the bid penalty by

$$\phi(B) = \frac{B^{\ln \pi}}{100}$$

Then the final score of a team is given by

$$F = P - \phi(B).$$

Teams are ranked within Infinite Auction by *higher* final score  $F$ . That ranking determines your placement  $p_4$ . In the case of a tie, teams are tiebroken by *lower* cumulative bid sum  $B$ , and if still tied, by a random tiebreaker.